	RV COLLEGE OF ENGINEERING® Department of Computer Science and Engineering		
Course :	**Network Programming and Security**	**Course Code**	**CS362IA**
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Project Title	Secure AI-Based Surveillance System with PAKE Authentication		
Batch			
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INTRODUCTION TO THE TOPIC:

This project introduces a Secure AI-Based Surveillance System designed to overcome the vulnerabilities of traditional monitoring, such as weak authentication and data interception. By integrating Password Authenticated Key Exchange (PAKE) and encrypted communication, the system ensures secure access without ever exposing user credentials.

The platform features role-based access control and leverages AI-driven analytics for real-time object and motion detection. By combining robust cybersecurity with intelligent monitoring, this solution provides a highly reliable and proactive security framework for offices, campuses, and smart cities.

OBJECTIVE:

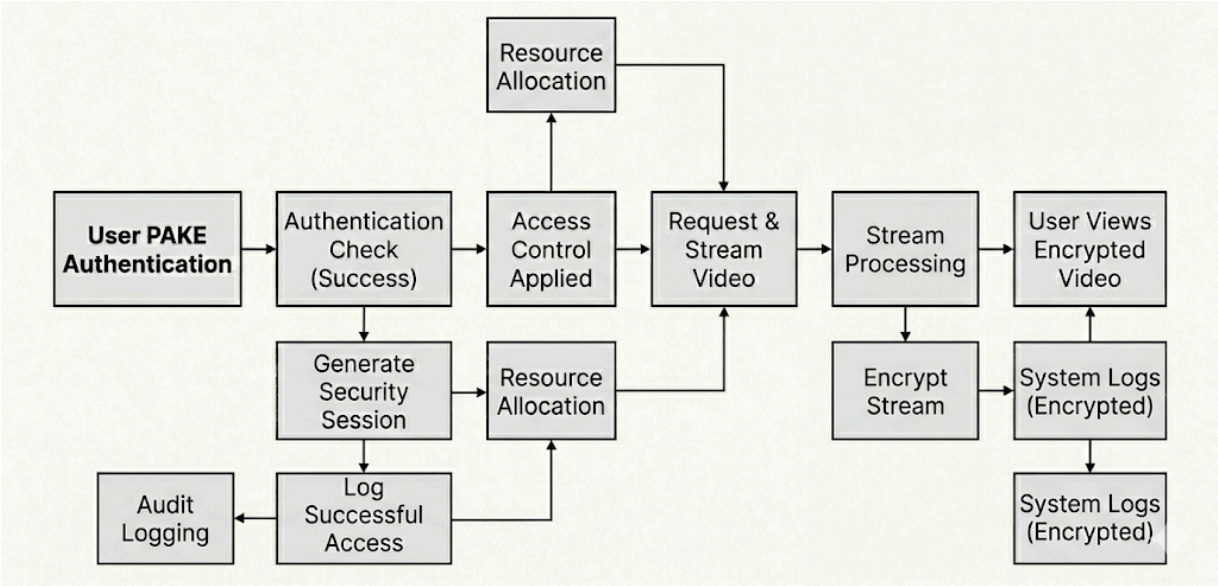
1. **To design a secure authentication system** using PAKE (Password Authenticated Key Exchange) that prevents password leakage during login.
2. **To implement role-based access control**, ensuring that users can only access authorized camera feeds.
3. **To develop a real-time camera streaming system** that allows users to monitor video feeds efficiently.
4. **To ensure secure data transmission** by encrypting video streams using protocols similar to VPN/TLS.
5. **To enhance system security and reliability** by preventing unauthorized access and protecting sensitive surveillance data.

METHODOLOGY:

Secure Authentication and Access Control

The system uses Password Authenticated Key Exchange (PAKE) to verify users without transmitting or storing passwords. After authentication, role-based access control ensures users only access permitted camera feeds, preventing unauthorized usage.

Secure Video Streaming and Processing
Camera feeds are streamed to the server, where AI techniques like motion or object detection are applied. The processed video is then securely transmitted to users over an encrypted channel, ensuring privacy and scalability**.**



INNOVATION / CONTRIBUTION IN NPS

The project integrates secure authentication using PAKE with encrypted video streaming and AI-based surveillance in a single system. It enhances security by preventing password exposure and restricting access through role-based control. The inclusion of AI enables intelligent monitoring instead of just passive viewing. Additionally, basic network simulation concepts such as multi-user access and latency handling are incorporated. This combination makes the system more secure, practical, and closer to real-world applications**.**

OUTCOME EXPECTED

- Expected OutcomesRobust Security: Implements PAKE-based authentication and end-to-end encryption to prevent credential exposure and data interception.
- Controlled Access: Utilizes Role-Based Access Control (RBAC) to manage multi-user permissions and restrict camera feeds to authorized personnel.
- System Integrity: Delivers a unified, scalable platform that ensures data confidentiality and proactive surveillance for real-world applications.

Signature of Lab Teacher